

In this documentation, I will record all the steps I go through in this project, as well as my thoughts on how I solve problems.

ETL Logic:

- **Data Exploration:** After I read the data, I create a for loop to show all info of data like (info, null. Duplicates) and I find there are 90% of rows in (Name, Education and Occupation) missing values . Its supposed I will drop these columns because the missing values are $\geq 60\%$, However, I can't delete those columns until I first go back to the business requirements and those columns are important, so I will replace the NULL with the value Unknown.
- **Data Transformation:** In this process, I begin by creating a copy of the original dataset to ensure that all transformations are performed on a separate working version, preserving the integrity of the source data. I then convert the *OrderDate* column to a proper Date data type to enable time-based analysis and extract a *Year* field, which is used to align and compare sales data with forecast data. Next, I handle data quality issues by replacing missing values with “Unknown” and removing duplicate records to ensure consistency and reliability. I also apply whitespace trimming across all columns to standardize text values and prevent mismatches when establishing relationships between tables. Finally, I create a new *Sales* column using the formula $Quantity \times Net_Price$ to calculate total sales value for each record. This structured approach ensures clean, analysis-ready data suitable for downstream reporting and modeling.
- **Data Loading:** Finally Once the data is cleaned and correctly linked through the appropriate columns, it is saved to support building a data model and enabling accurate reporting and visualization

Data Model Logic:

Now that the data is prepared, it is loaded to create a data model and establish relationships between datasets. A right outer join was used to link the tables based on common columns: *Country/Region*, *Brand*, and *Year*.

To ensure the relationships were correctly established, a validation check was performed. This involved grouping the data in *data_model* for the year 2009 (matching the forecast dataset) and comparing the aggregated results. After that, the columns *data_model.Forecast* and *forecast.Forecast* were compared to verify consistency. A validation function was created to check whether *Forecast* and *Forecast_true* values match. Any mismatch was flagged as a failure, indicating an inconsistency in the model.

key assumptions

1. Missing values:
I assumed missing categorical values represent unknown information and can be safely replaced with 'Unknown'
2. Dropping vs keeping columns :
I assumed these columns are business-important despite high missing rate.
3. Sales calculation
I assumed sales amount is correctly calculated by multiplying quantity by net price.
4. Date handling
I assumed OrderDate is the correct field for time-based analysis and yearly comparison.
5. Forecast alignment
I assumed forecast and sales can be compared using Year as the common granularity.